Game Design Doc - Part 1

Potionsmith

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One Page

Elevator Pitch

A single-player crafting and management simulation strategy game where you play as an alchemist, brewing potions, refining ingredients, and fulfilling customer requests to build your reputation and grow your potion shop.

Category

Single Player / Crafting / Shop Management / Time Management / Simulation / Strategy

Overview

Herbalism(working title) is a game built on the back of tying the game *Potion Craft* together with a more in-depth order fulfillment-based game, such as *Papa's Pizzeria* or *Papers Please*. The game places particular emphasis on tactile gameplay and explicit step following while using documents as its primary source of instruction. The player is set in the world of Ethshar, and this world is specifically emphasized to further develop a sense of immersion.

Play Value

The main play value of the game is the hands-on, exploratory experience that rewards curiosity and experimentation. By ordering, selecting, and refining ingredients, the player engages in tactile decision-making that directly impacts potion outcomes. Each ingredient carries unique properties, encouraging experimentation and strategic thinking when combining ingredients. This mechanic fosters a sense of discovery, mastery, and immersion, as the player gradually learns how different ingredients interact and influence the quality and effects of their potions. The game also connects the player to the world, providing a feedback loop between exploration, crafting, and the tangible impact of potions on the game world.

Vision Statement

Herbalism is a game about brewing potions for clients and placing the player into the shoes of an alchemist in the world of Ethshar.

Inspirational Media

Potion Craft

Genre:

Simulation / Crafting

Gameplay:

Players take on the role of an alchemist who experiments with various ingredients to brew potions. They manually grind herbs, mix liquids, control heating, and navigate a potion map to discover new recipes. The player interacts with customers, fulfilling potion requests to earn gold and reputation.

Resources:

- Ingredients (herbs, mushrooms, minerals)
- Water, oil, and other bases
- Gold (for trading and upgrades)
- Reputation and popularity (affecting customer trust and pricing)

Obstacles:

- Limited ingredient supply and gold
- Failed potion experiments waste materials
- Demanding or impatient customers
- Balancing moral choices (helping good or bad clients)

Play Value:

Potion Craft offers a relaxing, tactile crafting experience with a strong sense of discovery and experimentation. Its charm lies in the freedom to create, trade, and experiment at your own pace, rewarding curiosity and creativity.

Inspirations for Our Game:

- Potion crafting

Papa's Pizzeria

Genre:

Time Management / Simulation

Gameplay:

Players run a pizzeria where they take customer orders, prepare dough, add toppings, bake pizzas, and deliver them efficiently.

Resources:

- Ingredients (dough, sauce, cheese, toppings)
- Kitchen tools and ovens
- Money (to upgrade the pizzeria)
- Customer orders (as "requests")

Obstacles:

- Time pressure to fulfill multiple orders simultaneously
- Increasing complexity of customer requests
- Limited workspace and tools
- Balancing quality vs. speed

Play Value:

Fast-paced, satisfying, and addictive due to the combo of time management, multitasking, and visual feedback. Offers a sense of progression as players unlock upgrades, new ingredients, and more challenging customers.

Inspirations for Our Game:

- Order taking
- Customer orders
- Gameplay loop

Papers Please

Genre:

Puzzle / Simulation / Dystopian / Strategy

Gameplay:

Players take on the role of a border control officer in a fictional totalitarian country. They inspect travelers' documents, check for discrepancies, and decide whether to allow or deny entry.

Resources:

- Passports, ID cards, and other documents
- Rules and inspection tools (stamps, interrogation prompts, fingerprint checks)
- Money for the family (earned by correctly processing entrants)
- Information sheets (new regulations, directives, and updates)

Obstacles:

- Increasingly complex documents and rules
- Moral dilemmas (helping desperate people vs. following orders)
- Limited time to process travelers
- Penalties for mistakes, which can affect your salary and family welfare

Play Value:

Highly engaging for players who enjoy strategic thinking, pattern recognition, and moral decision-making. Creates tension through time pressure and consequences for choices, making each day feel meaningful.

Inspirations for Our Game:

- Scene layout design

Plants vs. Zombies

Genre:

Tower Defense / Strategy / Casual

Gameplay:

Players strategically place plants with different abilities along lanes to stop waves of incoming zombies from reaching their house.

Resources:

- Sun points (currency for planting)
- Various plant types (offensive, defensive, or utility)
- Garden space (limited slots per lane)

Obstacles:

- Waves of increasingly tough zombies with unique abilities
- Limited sun resources to deploy plants
- Space constraints in the garden lanes
- Timing and placement decisions under pressure

Play Value:

Engaging for both casual and strategic players. Offers a satisfying mix of planning, quick thinking, and progression with new plants and zombie types. Its charm comes from humorous visuals and escalating challenge.

Inspirations for Our Game:

- Seed selection phase before the combat

Good Pizza, Great Pizza

Genre:

Time Management / Simulation / Casual

Gameplay:

Players run a pizza shop, taking customer orders and assembling pizzas according to specific requests. Gameplay involves preparing dough, adding toppings, baking, and delivering pizzas while managing customer satisfaction.

Resources:

- Ingredients (dough, sauce, cheese, toppings)
- Kitchen tools and ovens
- Money (to buy ingredients and upgrade the shop)
- Customer orders (as “requests”)

Obstacles:

- Increasingly complex customer orders
- Time constraints to complete orders
- Balancing quality vs. speed for customer satisfaction
- Managing finances and limited resources

Play Value:

Fun, fast-paced, and addictive for players who enjoy multitasking and time management. Offers a satisfying loop of fulfilling orders, earning money, and upgrading the shop, combined with colorful visuals and humor.

Inspirations for Our Game:

- Customer UI
- Time constraint
- End of the day report

Market Analysis

Audience

People who want to feel immersed in the role of an alchemist, as well as people who like time crunch, or management games such as the Papa's series.

Platform

PC. The demand for mouse controls makes this platform the easiest to develop this style of game for.

Competition Research

The biggest competition would be *Potion Craft* itself. Though potion games do exist, they have a nasty habit of feeling intangible, so one goal would be to see or get to interact with the effects of the potions themselves.

Project Plan

Scope

For the final product, the project aims to consist of a tutorial and at least one level. This includes full mechanical interactions with each machine as well as utility for all included ingredients. For the final version, we need at least 5 refinements and 4 ingredients, and at least 3 potion effects with an amplified version for each. At least two customers, one being the tutorial, one being the level.

Timeline

Sprint 1

Design: Brainstorm the game idea, determine gameplay elements, and design mechanics.

Sprint 2 (MVI)

Implementation: Make a potion, refine an ingredient, and confirm a sale.

Aesthetics: Using temporary assets, but finalize the art style, create concept art, and design the scene layout.

Sprint 3

Design: Minor pivot in gameplay loop design (from multiple customers per day(level) to focusing on one per day(level)).

Implementation: Additional refinement options and a single potion with multiple steps, add customer interactions, and a dialogue system.

Aesthetics: Create assets for ingredients and tools, and start to replace the temporary assets in the scene.

Sprint 4

Implementation: Implement potion effects and their amplified versions, and start tracking potion outcomes.

Aesthetics: Design UI elements for potion brewing, customer requests, and dialogue windows; improve environment visuals.

Sprint 5

Implementation: Integrate multiple ingredients and refinement combinations, implement the tutorial customer flow, and finalize core game logic.

Aesthetics: Replace remaining placeholder assets, polish transitions, and add sound effects for mixing, refining, and selling.

Sprint 6

Implementation: Add the second (main) customer level with unique potion requests, refine balancing, and finalize all mechanical interactions.

Aesthetics: Final background details; ensure cohesive art direction and consistency across scenes.

Final

Implementation: Conduct full integration testing, fix bugs, and ensure all systems (crafting, selling, dialogue, progression) function smoothly.

Aesthetics: Apply final polish—refine UI, adjust colors and effects, and prepare the build for presentation.

Team Roles

Producer - Samuel Denton

Artist - Brady Marr

Main Gameplay Programmer - Kevin Zheng

Assistant Programmer/Bugfixing - Galen Sagarin

Core Gameplay

Core Mechanic

The core game mechanic centers around potion making, which integrates several sub-mechanics such as ingredient picking, refining, and bottling. The player will craft potions by selecting from a set of available ingredients, which are gradually unlocked as the game progresses. Each ingredient has a unique effect, and players must carefully combine ingredients to fulfill specific potion requests. The completion of refinements affects the quality of the ingredients, which affects the quality of the final product, the potion.

Core Gameplay Loop

Each day begins with the town crier announcing the news, providing hints about upcoming customer needs. The player then proceeds to order ingredients and prepare the shop for business.

The core of the game revolves around a simple yet engaging loop of potion crafting and delivery. Players start by receiving orders and identifying the desired effects for each potion. They then refine ingredients through tactile mechanics such as grinding and plucking to prepare them for brewing. Once ready, ingredients are added to the cauldron, where they are combined and mixed to create the potion.

After brewing, the finished potion is handed to the customer by placing it in the designated delivery area. At the end of the day, a summary report highlights earnings, reputation changes, and customer feedback, giving players insight into their overall progress and performance.

Secondary Mechanics

Order System

Customer Requests are delivered through dialogue between the customer and the player. During the interaction, the customer explains the specific effects they need from the potion.

Order Fulfillment occurs when a player completes a customer's order by delivering a crafted potion. This is accomplished through a drag-and-drop interaction, where the player picks up

the crafted potion and places it on the designated area of the counter. Successfully delivering the potion completes the order, triggering updates to the customer's feedback, earnings, and reputation.

Rating is the assessment of how well a product fulfills a customer's request. Under the hood, the system compares the final effects of the brewed potion to the customer's stated requirements.

- For a single effect, the formula is:
    - rating = effectiveness_in_potion / needed_effectiveness
- For multiple effects, the process is:
    - Calculate each (effectiveness_in_potion / needed_effectiveness) * (1 / number_of_needs)
    - Add the results together
    - All ratings will be clamped to 1, where 1 represents 100% fulfillment of the requirements.

End of the Day Report is the closing scene of each level, summarizing the player's performance for that day. The report displays key metrics such as total profit and loss, overall store reputation, and individual reputation with special customers.

Town Crier System

Before the Ingredient Ordering phase, a town crier enters the shop and informs the player about recent events in the game world. This serves as a **direct feedback system**, showing the player how the potions they created have impacted the world. Furthermore, it provides an opportunity for the player to learn about current trends and better prepare for upcoming orders.

Resource Management

Time Constraint requires the player to complete the order within a set time frame. If the player fails to finish the potion before time runs out, the customer will cancel the order, resulting in lost potential earnings and reputation. Conversely, completing the order within the time limit rewards the player with a high reputation.